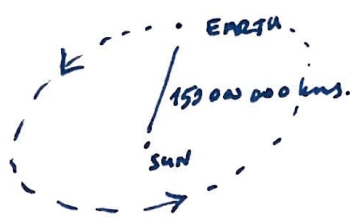


Location & location

Notion of a particle: A body whose dimensions may be ~~neglected~~ ^{reflected} describing its motion. For example, the planets may be regarded as particles in considering their motion about the sun. But when it comes to explain, for example, tidal waves or tornadoes we cannot neglect ~~Earth's~~ Earth's ~~rotation~~ rotation about its axis.

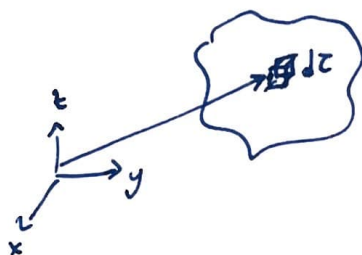


→ A point in space (no dimensions) with an associated mass.

A problem: density $\rho(x, y, z) = \begin{cases} \infty & \text{at } (0, 0, 0) \\ 0 & \text{at elsewhere} \end{cases}$

(?) How are we going to describe the total mass of this point with maths if we are only given in terms of ρ ?

If we had a mass distribution

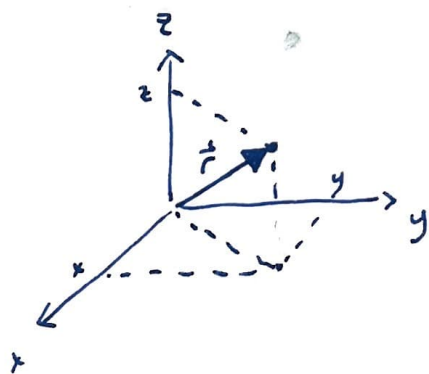


$$\int \rho(x, y, z) dV = M$$

Can we apply this for a point particle as well?

$\int \rightarrow \infty$
 $M \rightarrow \text{finite}$ } what is going on?

(1) Location & location How do we locate a particle in 3D space?



We will describe the position of a particle with a vector called the position vector.

$$\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$$

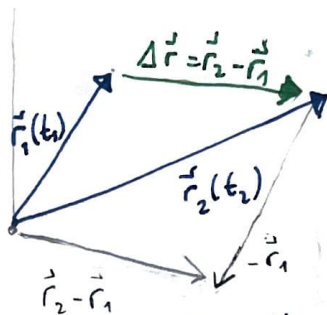
More generally ^{functions of time.}

$$\vec{r}(t) = x(t)\hat{i} + y(t)\hat{j} + z(t)\hat{k}$$

→ position vector.

(2) How do we describe the motion of a particle moving from a position $\vec{r}_1(t_1)$ to another position $\vec{r}_2(t_2)$?

→ Its motion can be described by another vector called the displacement vector:



$$\Delta \vec{r} = \vec{r}_2(t_2) - \vec{r}_1(t_1) : \text{displacement vector}$$

→ the vector particle is displaced this much starting from the position $\vec{r}_1(t_1)$

③ How are we going to describe the velocity of the particle then?



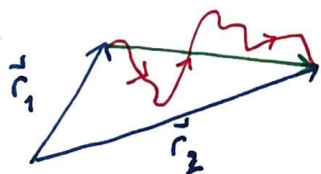
~~the particle has displacement~~

• Velocity is defined as the change in position with respect to time. $\frac{\Delta \vec{r}}{\Delta t} = \vec{v}$

If the displacement of the particle is described by $\Delta \vec{r}$, can we define its "velocity" (I mean time rate of change of its position) as

$$\vec{v} \stackrel{?}{=} \frac{\Delta \vec{r}}{\Delta t} : \text{what does this describe.}$$

Maybe the particle moved between the time interval $t_1 - t_2$ as



Maybe it sped up during its motion or slowed down or maybe it took a brake for some time. we don't know. The information given to us in the beginning was only $\vec{r}_1(t_1)$ and $\vec{r}_2(t_2)$. Therefore all we can say is how fast and in which direction the particle moved during Δt on average.

$$\vec{v}_{avg} = \frac{\Delta \vec{r}}{\Delta t} : \text{same direction as the displacement vector.}$$

④ How does $\vec{v}_{avg}$ look like in component form given $\vec{r}_1$ and $\vec{r}_2$?

$$\vec{r}_1(t_1) = x_1(t_1)\hat{i} + y_1(t_1)\hat{j} + z_1(t_1)\hat{k}$$

$$\vec{r}_2(t_2) = x_2(t_2)\hat{i} + y_2(t_2)\hat{j} + z_2(t_2)\hat{k}$$

$$\Delta \vec{r} = \vec{r}_2 - \vec{r}_1 = [x_2(t_2) - x_1(t_1)]\hat{i} + [y_2(t_2) - y_1(t_1)]\hat{j} + [z_2(t_2) - z_1(t_1)]\hat{k}$$

$$\vec{v}_{avg} = \frac{x_2 - x_1}{\Delta t}\hat{i} + \frac{y_2 - y_1}{\Delta t}\hat{j} + \frac{z_2 - z_1}{\Delta t}\hat{k}$$

⑤ what if the velocity of the particle is changing in each moment of time and we are interested in the velocity of the particle at a certain

$$\lim_{\Delta t \rightarrow 0} \vec{V}_{avg} = \lim_{\Delta t \rightarrow 0} \frac{x_2(t_2) - x_1(t_1)}{\Delta t} \hat{i} + \dots \hat{j} + \dots \hat{k}$$

Therefore, the velocity of the particle at instant t can be given:

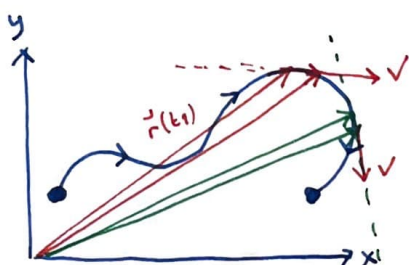
$$\vec{V}(t) = \frac{dx}{dt} \hat{i} + \frac{dy}{dt} \hat{j} + \frac{dz}{dt} \hat{k} = V_x(t) \hat{i} + V_y(t) \hat{j} + V_z(t) \hat{k}$$

$\downarrow$ velocity in x direction $\downarrow$ velocity in y direction $\downarrow$ velocity in z direction.

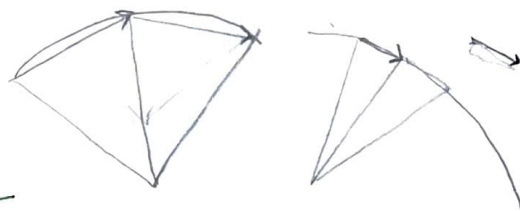
Speed of the particle at time t is defined as

$$v(t) = |\vec{V}(t)|$$

- ⑥ How would the displacement and velocity vectors ~~look like~~ ^{would be directed} on the trajectory of a particle?



Velocity and displacement vectors are always tangential to the particle's trajectory



- ⑦ How do we describe acceleration ~~units~~ units?

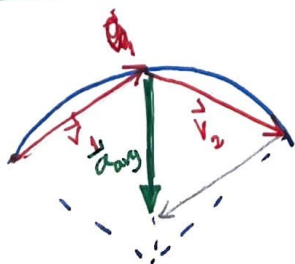
Acceleration: Time rate of change of velocity; $\frac{[L]}{[T]^2} = \frac{m}{s^2}$

$$\vec{a}(t) = \frac{d}{dt} [\vec{V}(t)] = \frac{d}{dt} [V_x(t) \hat{i} + V_y(t) \hat{j} + V_z(t) \hat{k}] = \frac{d}{dt} [V_x(t)] \hat{i} + \frac{d}{dt} [V_y(t)] \hat{j} + \frac{d}{dt} [V_z(t)] \hat{k}$$

$$= a_x(t) \hat{i} + a_y(t) \hat{j} + a_z(t) \hat{k}$$

accelerations in respective directions.

- ⑧ What would be the direction of ^{vector} acceleration ~~average~~ if a particle follows a curve with a constant radius?



$$\vec{a}_{avg} = \frac{\Delta \vec{V}}{\Delta t} = \frac{\vec{V}_2 - \vec{V}_1}{\Delta t}$$

Summary: Once we are given $\vec{r}(t)$ for a particle, it means that we know everything about its motion:

$$\vec{r}(t) \xrightarrow{d/dt} \vec{V}(t) \xrightarrow{d/dt} \vec{a}(t)$$

⑨ Newton's notation for derivatives

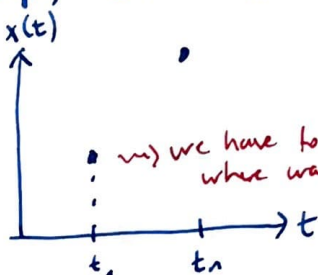
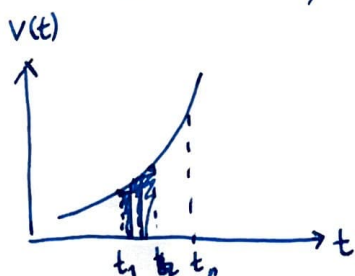
$$\frac{d}{dt} [\vec{r}(t)] \equiv \dot{\vec{r}}(t) = \vec{v}(t)$$

$$\frac{d}{dt} \frac{d}{dt} [\vec{r}(t)] \equiv \ddot{\vec{r}}(t) = \vec{a}(t)$$

$$\frac{d}{dt} [\vec{v}(t)] \equiv \dot{\vec{v}}(t) = \vec{a}(t)$$

⑩ We know what to do to obtain $\vec{v}(t)$ and $\vec{a}(t)$ once we are given $\vec{r}(t)$.
Is it possible to go the other way around?

i.e.) If we are given $\vec{v}(t)$, for example, can we find $\vec{r}(t)$?



we have to know this
where was the particle at time t_1 ?

$$\Delta x = \int_{t_1}^{t_2} v_x(t) dt$$

If we are given the form of $v_x(t)$, we can find the form of $x(t)$
up to a constant

$$x(t) = \int v_x(t) dt + C$$

Similarly

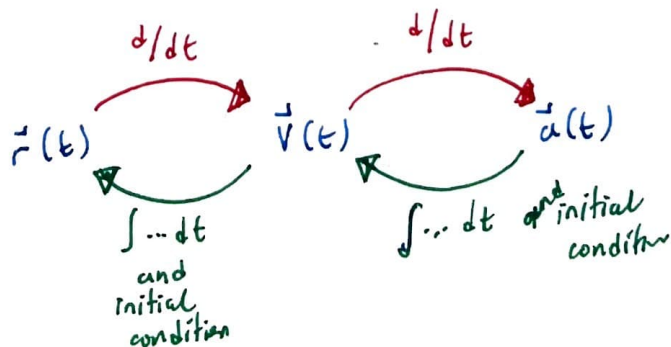
$$v(t) = \int a_x(t) dt + D$$

In general:

$$\vec{v}(t) = \int \vec{a}(t) dt + C_1$$

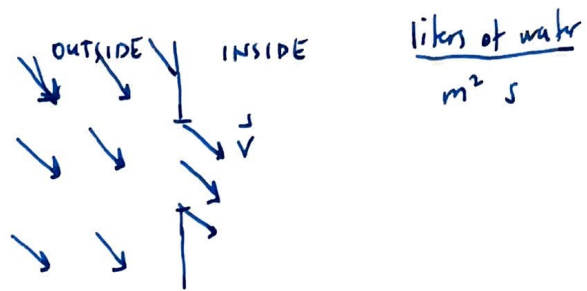
$$\vec{r}(t) = \int \vec{v}(t) dt + C_2$$

Summary:

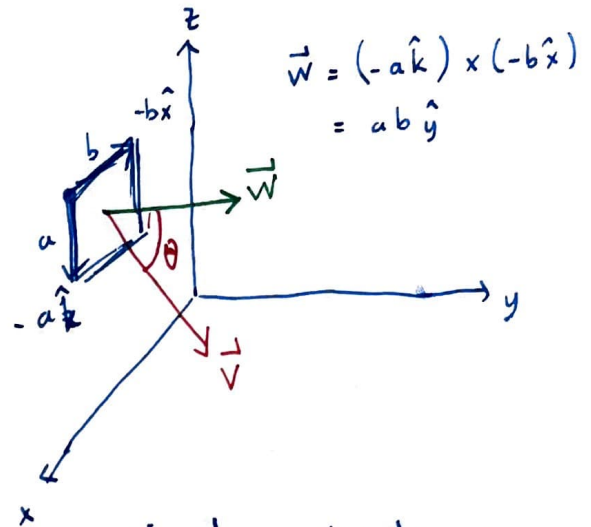
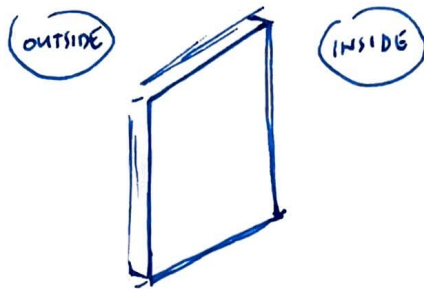


Ex: Rain getting into the room.
 Flux $\rightarrow$ Let vector $\vec{B}$ & $\vec{V}$ represent the flux of rain

Flux: amount of sth
 $[L]^2 [T]$



liters of water
 $m^2 s$



$$\vec{W} = (-a\hat{k}) \times (-b\hat{x}) = ab\hat{y}$$

$$\vec{V} \cdot \vec{W} = |\vec{V}| |\vec{W}| \cos \theta$$

Kinematics (Examples for types of motion)

Linear motion : Motion taking place on a line.

$$\begin{aligned} x(t) &= at \\ y(t) &= bt \\ z(t) &= ct \end{aligned} \quad \vec{r}(t) = at\hat{i} + bt\hat{j} + ct\hat{k} = \begin{pmatrix} at \\ bt \\ ct \end{pmatrix} : 3D \text{ motion}$$

Ex: If we have 2 particles moving on a plane. what time do they collide and where?

$$\begin{aligned} \vec{r}_1(t) &= t\hat{i} + t\hat{j} \\ \vec{r}_2(t) &= (-t+4)\hat{i} + (2t-2)\hat{j} \end{aligned} \quad \left. \begin{array}{l} \\ \end{array} \right\} \xrightarrow{d/dt} \begin{aligned} \vec{v}_1(t) &= \hat{i} + \hat{j} \\ \vec{v}_2(t) &= -\hat{i} + 2\hat{j} \end{aligned}$$

Define separation vector $\Delta\vec{r}$

$$\Delta\vec{r} = \vec{r}_2(t) - \vec{r}_1(t) = (-2t+4)\hat{i} + (t-2)\hat{j}$$

$$|\Delta\vec{r}| = 0 \quad \Rightarrow \quad \sqrt{(-2t+4)^2 + (t-2)^2} = 0$$

$$\sqrt{4t^2 - 16t + 16 + t^2 - 4t + 4} = 0$$

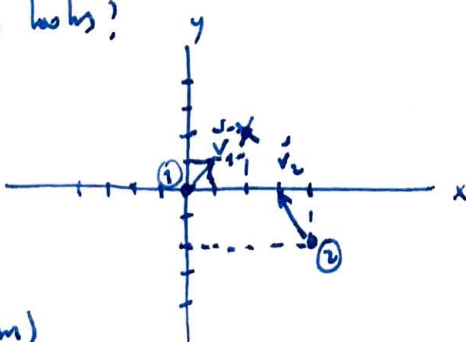
$$\sqrt{5t^2 - 20t + 20} = 0 \quad \sqrt{t^2 - 4t + 4} = 0$$

$$\sqrt{(t-2)^2} = 0$$

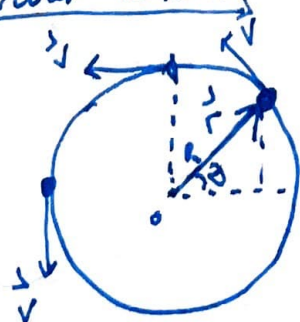
Where? $\vec{r}_1(2) = 2\hat{i} + 2\hat{j}$

$$t-2 = 0 \rightarrow t=2$$

How it looks?



(uniform)
Circular motion:



Let's start by describing its position

$$\vec{r}(t) = R \cos \theta \hat{i} + R \sin \theta \hat{j}$$

but we know $\vec{r}$ changes with time.

function of θ .

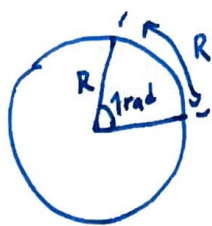
in radians (unitless)

$$\theta(t) = \omega t$$

rad/s

→ discussion about "unit" radians

What is radian?



One radian is defined as the angle at the center of a circle in a plane that corresponds to an arc whose length equals the radius of the circle.

$$R \rightarrow 1 \text{ rad}$$

circumference of circle $(2\pi R) \rightarrow 2\pi \text{ rad}$

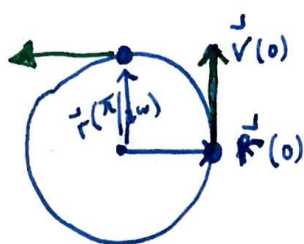
$\theta(t) = \omega t$: definition of angular velocity.

How fast ^{does} that angle θ change with respect to time

$$\vec{r}(t) = R \cos(\omega t) \hat{i} + R \sin(\omega t) \hat{j}$$

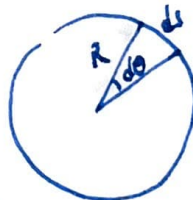
$$\vec{v}(t) = -R\omega \sin(\omega t) \hat{i} + R\omega \cos(\omega t) \hat{j}$$

$$\vec{v}(t) = -R\omega \sin(\omega t) \hat{i} + R\omega \cos(\omega t) \hat{j} \rightarrow \text{does it make sense?}$$



$v = \omega R \rightarrow$ Is there any other way to make this legit?

$$R \rightarrow 1 \text{ rad do}$$



$$ds = R d\theta$$



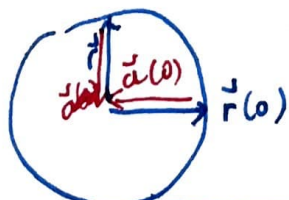
det of radian

$$s = R\theta$$

$$\frac{ds}{dt} = R \frac{d\theta}{dt} = R\omega$$

$$\vec{a}(t) = -R\omega^2 \cos(\omega t) \hat{i} - R\omega^2 \sin(\omega t) \hat{j}$$

$\rightarrow$ Does it make sense?



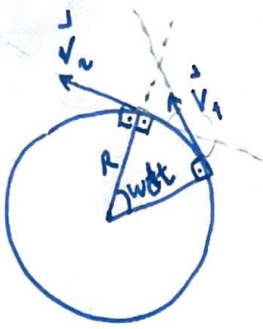
$$|\vec{a}(t)| = \sqrt{R^2 \omega^4 \cos^2(\omega t) + R^2 \omega^4 \sin^2(\omega t)}$$

$$= \sqrt{(R\omega^2)^2 [\cos^2(\omega t) + \sin^2(\omega t)]}$$

$$= R\omega^2 = a = R \frac{v^2}{R^2} = \frac{v^2}{R}$$

$\rightarrow$ Is there a way to show this by simply using vector addition and geometry?

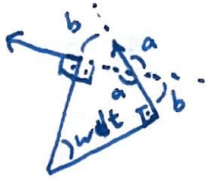
Yes.



If we know this angle we can find the magnitude of c which will give us the magnitude of acceleration

$$-\vec{v}_1, \vec{v}_2 - \vec{v}_1 = d\vec{c}$$

$$v^2 + v^2 - 2v^2 \sin wdt = c^2$$



$$a = 180 - wdt$$

$$2(180 - wdt) + 2b = 360$$

$$b = wdt$$

$$dc = v wdt$$

$$= wR wdt$$

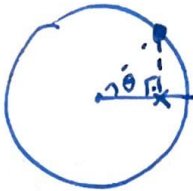
$$acc \left(\frac{dc}{dt} \right) = w^2 R = \frac{v^2}{R}$$

Period:

$$T = \frac{2\pi R}{v}$$

$$= \frac{2\pi R}{wR} = \frac{2\pi}{w}$$

Oscillatory motion

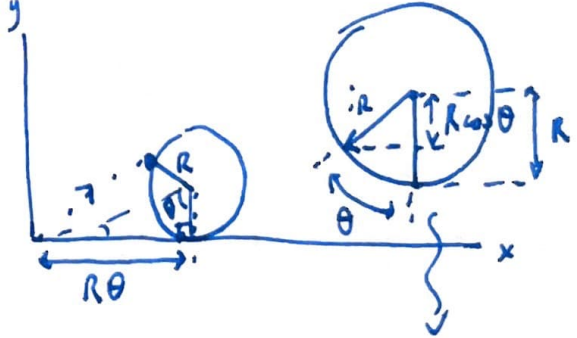
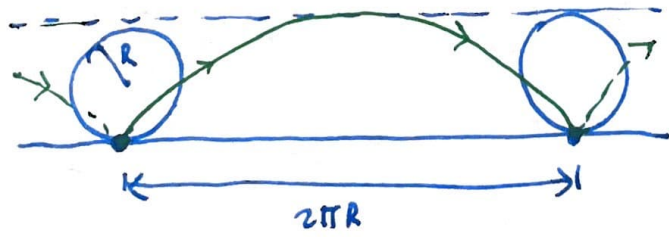


$$x(t) = w s w t$$

we will ~~to~~ cover this later.

Cycloidal motion

It is the path traced by a point on the circumference of a circle as it rolls along a straight line without slipping. (Imagine drawing a chalk dot on a bicycle tire and rolling the bicycle forward - dot traces a cycloid.)



$$\vec{r}_1(\theta) = R\theta \hat{i}$$

$$y(t) = R [1 - \cos(wt)]$$

$$x(t) = R [wt - \sin(wt)]$$

$$\vec{r}(t) = x(t) \hat{i} + y(t) \hat{j}$$

$$y(\theta) = R - R \cos(\theta)$$

$$x(\theta) = R\theta - R \sin(\theta)$$

Brachistochrone.
curve is
cycloid.